

Matgeo-q.8.3.16

AI25BTECH11036-SNEHAMRUDULA

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Question. An arch is in the form of a semi-ellipse. It is 8 m wide and 2 m high at the centre. Find the height of the arch at a point 1.5 m from one end.

solution

solution. We are given a semi-ellipse of width 8 m and height 2 m at the centre.

For an ellipse, the equation is

$$\mathbf{q}^T \mathbf{V} \mathbf{q} = 1 \quad (1)$$

where

$$\mathbf{V} = \begin{pmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{b^2} \end{pmatrix} \quad (2)$$

Here, the semi-major axis $a = 4$ (half of 8 m) and semi-minor axis $b = 2$.
Thus,

$$\mathbf{V} = \begin{pmatrix} \frac{1}{16} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \quad (3)$$

Let

$$\mathbf{q} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (4)$$

Substituting into (1),

$$\frac{\alpha^2}{16} + \frac{\beta^2}{4} = 1 \quad (5)$$

Since the width is 8 *m*, the end corresponds to $\alpha = \pm 4$. At a point 1.5 *m* from the end, the coordinate is

$$\alpha = 4 - 1.5 = 2.5 \quad (6)$$

Substitute $\alpha = 2.5$ in (6):

$$\frac{(2.5)^2}{16} + \frac{\beta^2}{4} = 1 \quad (7)$$

$$\frac{6.25}{16} + \frac{\beta^2}{4} = 1 \quad (8)$$

$$\frac{\beta^2}{4} = 1 - 0.390625 \quad (9)$$

$$\frac{\beta^2}{4} = 0.609375 \quad (10)$$

$$\beta^2 = 2.4375 \quad (11)$$

$$\beta \approx 1.561 \quad (12)$$

Hence, the height of the arch at a point 1.5 m from the end is approximately

$$\beta \approx 1.56 \text{ m} \quad (13)$$

Graphical Representation

